

Chapter 16

Phonosemantic conditioning in Kirundi loanword noun class allocation

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This paper examines the phonological and semantic factors in the noun class allocation of loanwords in Kirundi. I provide a formal account of the criteria used in determining the noun class of loanwords during the lexicalization process. The data indicates an interaction between phonological and semantic factors. I formalize the relevant factors as Optimality Theory constraints and describe a ranking capable of predicting the noun class allocation of novel loanwords in Kirundi. The analysis taken here calls for greater unity between the disciplines of semantics and phonology, traditionally taken to be distinct save for certain phenomena, such as prosody.

1 Introduction

This paper examines the noun class allocation of loanwords in Kirundi, a Bantu language of Burundi, providing a semantic and phonological account of the noun class loanwords are assigned. Kirundi employs a noun class system expressed through noun class prefixes. Noun class pairings are pairs of noun classes that nouns typically alternate between to express number. Generally, Kirundi loanwords appear in class 9/10 (Zorc & Nibagwire 2007, Niyondagara 1993, Niyonkuru 1987). However, there are several other semantic and phonologically motivated factors in determining noun class. I present examples where semantic factors override phonological criteria. Restrictions on semantic content within a noun class may block nouns whose phonological form would otherwise condition



them to appear with a different prefix. I demonstrate instances where phonological factors place nouns in noun classes inconsistent with their semantic properties. Neither phonological, perceptive, nor semantic factors solely account for the allocation of noun class in Kirundi loanwords. The data indicate a clear interaction between phonological and semantic constraints on faithfulness and markedness. A formal ranking is provided, capable of predicting the noun class allocation of novel loanwords in Kirundi.

Section 2 provides a brief background of Kirundi's noun class system. Section 3 presents data on loanwords in Kirundi and identifies core morphological and phonological factors in determining noun class of Kirundi loanwords. In Section 4, I formalize these morphophonological factors in an Optimality Theory (OT) framework by supplying constraints and justifying their ranking. Section 5 presents apparent irregularities in allocating Kirundi loanwords to the noun class system, where semantic aspects of noun classes affect phonological principles. Section 6 accounts for semantic effects by providing formal semantic constraints ranked critically with morphophonological constraints discussed in Section 4. I show a clear interaction between semantic, morphological, and phonological constraints and advocate for a single merged interface accounting for noun class allocation, the implications of which are discussed in Section 7. In Section 8, I summarize the findings and provide a conclusion.

2 The noun class system

A prominent feature of many Bantu languages is their noun class system. Van de Velde (2019) defines the Bantu noun class system as a grouping of nouns which trigger the same agreement pattern. Bantu noun classes are often seen as analogous to grammatical gender. Maho (2009) notes that, on average, most Bantu languages have about fifteen noun classes. This number excludes locative noun classes, generally glossed as any noun class greater than 15. I do not consider locative noun classes in my analysis.

In Kirundi, noun classes are realized overtly by a noun class prefix. These appear on most nouns, although some proper and human-denoting nouns may appear without the prefix. The Kirundi noun class prefixes are shown in Table 1, where single noun classes appear on the left and plural/ non-individuated prefixes appear on the right¹. I display the augment in brackets in Table 1 to capture

¹Note that I use *n* to refer to a homorganic nasal unspecified for place of articulation. This is either realized as prenasalization or a nasal stop whose place is determined by the following vowel.

that it does not appear in all environments, discussed below in Section 2.1.² I omit the augment from later discussion for sake of clarity.³ Nouns are often inherently associated with a particular noun class but may shift for derivational purposes (Givón 1972; as cited in Aunio 2006). I later discuss this in Section 5.3.

Table 1: Kirundi noun class prefixes

Class	Prefix	Class	Prefix
1	(u)mu-	2	(a)βa-
3	(u)mu-	4	(i)mi-
5	(i)Ø / (i)ri-	6	(a)ma-
7	(i)ki- / (i)gi-	8	(i)βi-
9	(i)N- / (i)Ø-		
10	(i)N- / (i)Ø-		
11	(u)ru-		
12	(a)ka- / (a)ga-	13	(u)tu- / (u)du-
14	(u)βu-		

Kirundi expresses number opposition through noun class shift. Odd-numbered noun class prefixes generally refer to singular entities, while corresponding even-numbered noun class prefixes express plurality (Ndayiragije et al. 2012, Van de Velde 2019, Zorc & Nibagwire 2007: , a.o.)⁴. In (1), we see an alternation between singular and plural interpretations of the noun with class 1 and class 2 prefixes. In the proceeding sections, I refer to individual noun classes simply as ‘class 1’, for example, and the corresponding noun class pairing as ‘class 1/2’.

²As noted by an anonymous reviewer, the appearance or lack of the augment is conditioned by the syntactic and/or phonological environment it appears in. While the augment may not be truly ‘optional’ in that nouns in certain environments may appear augmentless, the crucial consideration for my analysis is that the augment is not a part of the noun in the same way as noun class prefixes. All nouns in Kirundi surface bearing a noun class prefix (though it may be covert) regardless of their syntactic environment. The same cannot be said for the augment, as there are certain environments where nouns must be augmentless. I refer the reader to Ndayiragije et al. (2012) and Shanks (2023) for in-depth discussions on factors conditioning the appearance of the augment in Kirundi.

³While some may object to the omission of the augment from later discussion, the fact that the augment is conditioned by its syntactic environment lead me to assume it is not a fundamental aspect of nouns in the same way as noun class prefixes.

⁴For an alternative analysis, see Mufwene (1980).

- (1) a. u-mu-gaβo
A-CL1-man
'man'
b. a-βa-gaβo
A-CL2-man
'men'

2.1 The augment

The last relevant piece of background for Bantu noun class systems is the augment. The exact nature of the augment in Bantu languages is contentious, and Van de Velde (2019) claims the role and function of the augment must be evaluated on the basis of individual Bantu languages. All that must be said of the augment now is that its form is of an initial vowel appearing before the noun class prefix on nouns. The augment's realization in Kirundi is determined by the following noun class prefix, copying the vowel quality of the vowel in the noun class prefix. The sole exception is in the class 9/10 prefixes, which appear either as prenasalization or as a null element. In this case, the form of the augment is [i].

With this background being established, the next section describes the set of phonological generalizations regarding noun class allocation for loanwords. I first present the clear phonological patterns in Section 3 and propose an initial analysis for them in Section 4 before moving on to discuss the interaction of semantic factors in Section 5.

3 Phonological conditioning in noun class allocation

Most Kirundi loanwords are lexicalized in class 9/10, as seen in the examples in Table 2 below. For clarity, I present all forms without the augment. I assume that the augment is added subsequent to noun class prefixes in the derivation.

An analysis of Kirundi loanword allocation would be quite simple if loanword nouns always appear in class 9/10. There are exceptions to this pattern, however. The first situation is loanwords with initial segments resembling existing noun class prefixes in Kirundi. In such cases, the initial segment sequence in the loanword is "reanalyzed" as a noun class prefix, allocating the loanword to the corresponding noun class (Niyondagara 1993, Niyonkuru 1987). Table 3 shows examples of this, where the initial segments are repurposed.

Table 2: Kirundi loanwords in class 9/10

	Kirundi	Gloss	Source
a.	Ø ₉ -mafi:ne	‘machine’	French, /mafɪn/
b.	Ø ₉ -sukari	‘sugar’	French, /sykr/
c.	Ø ₉ -farine	‘flour’	French, /farin/
d.	Ø ₉ -fero	‘iron’	French, /fɛr/
e.	Ø ₉ -kurudzeti	‘zucchini’	French, /kurʒɛt/
f.	Ø ₉ -ko ⁿ dʒe	vacation	French, /kõʒe/

Table 3: Reanalysis of initial segment as noun class prefix

	Kirundi	Gloss	Source
a.	mu ₃ -sigiti	‘mosque’	Arabic, /masdʒid/
b.	ma ₆ -joneze	‘mayonnaise’	French, /mɛʒonɛs/
c.	ki ₇ -ro	‘kilogram’	French, /kilo/
d.	βu ₁₄ -re ⁿ geti	‘blanket’	English, /bleɪnkɪt/
e.	ki ₇ -nini	‘pill’	French, /kinin/

Kirundi also disallows consonant clusters and has several strategies for resolving them in loanwords, such as vowel epenthesis. Different criteria may determine the form of the epenthesized vowel. In some cases, the epenthesized vowel is a copy of an adjacent vowel or is derived from place features of the preceding consonant. However, we see instances of the epenthesized vowel differing from what’s expected if another vowel could form a recognizable noun class prefix. I provide a deeper analysis of epenthetic vowel forms in Section 4.3.

The reader may immediately notice the apparent disparity between the forms (a) and (b) in Table (3). Both source words begin with the same CV sequence, yet only (b) in Table 3 allows that sequence to be repurposed as a noun class prefix. I will argue that it is the different semantic categories of the two nouns that allow repurposing of (b) but not (a). Specifically, I argue in Section 5.2 that *ma₆-joneze* is realized as class 6 due to its having cumulative reference, while *Ø-mafi:ne* is blocked from class 6 for its lack of cumulative reference.

In the proceeding section, I argue that a simple ranking of alignment constraints and constraints on the realization of morphemes can account for this described behaviour.

4 A formal account of phonological factors

4.1 Spellout and boundaries

An observed commonality between the nouns in Table 2 and those in Table 3 has to do with word boundaries. The word-initial segments in all examples from the source language's input maintain their position as left-aligned to word boundaries in the loanword, only preceded by the augment. Compare the forms in Table 2 and Table 3 with the ill-formed candidates in Table 4.

Table 4: Ill-formed candidates

	Kirundi	Gloss	Source
a.	*mu ₃ -mafi:ne	'machine'	French, /mafɪn/
b.	*mu ₃ -sukari	'sugar'	French, /sykr/
c.	*ki ₇ -farine	'flour'	French, /farin/
d.	*ki ₇ -fero	'iron'	French, /fɛr/
e.	*ki ₇ -kurudzeti	'zucchini'	French, /kurʒɛt/
f.	*mu ₃ -ko ⁿ dʒe	vacation	French, /kõdʒe/

The ill-formed candidates in Table 4 have extra phonological material between the source word's input and the word edge in the form of an overt noun class prefix. I posit the ill-formedness of these candidates is explained as the violation of an alignment constraint maintaining word boundaries. This constraint is formalized as:

(2) **FAITH-ALIGN-SB**

Maintain word boundaries from the source language in the base language (i.e. the receiving language)⁵

This simple constraint predicts which class most loanwords fall. Class 9/10 is the most common noun class for loanwords, as its prefix is phonologically null in most cases. Class 9/10's null prefixes avoid violation of FAITH-ALIGN-SB while ensuring the noun still receives proper nominal morphology. FAITH-ALIGN-SB

⁵An anonymous reviewer asks if repurposing forms such as those in Table 3 violate FAITH-ALIGN-SB. For the alignment constraint in (2) to work as intended, there is a necessary assumption that Kirundi nominals only form a complete prosodic word when prefixed with noun class prefixes. Thus, the forms in Table 3 would constitute a violation of FAITH-ALIGN-SB only if the root were a prosodic unit. However, I assume that FAITH-ALIGN-SB is evaluated of a prosodic word with the noun class prefix already attached.

explains the relative prominence of loanword allocation to class 9/10 when no existing segments are available for repurposing.

This constraint alone cannot account for the forms seen in Table 3. It would certainly be notably easier to assign all loanwords to a single noun class. Thus, it must be explained why existing segments are reinterpreted as class prefixes when applicable rather than simply assigning every loanword to class 9/10. The analysis should account for the ill-formedness of Table 5 compared to Table 3 for not effectively utilizing the existing phonological material given by the source language as a prefix.

Table 5: Ill-formed candidates

	Kirundi	Gloss	Source
a.	*Ø ₉ -musigiti	‘mosque’	Arabic, /masdʒid/
b.	*Ø ₉ -majoneze	‘mayonnaise’	French, /meʝones/
c.	*Ø ₉ -kiro	‘kilogram’	French, /kilo/
d.	*Ø ₉ -βure ⁿ geti	‘blanket’	English, /blemkɪt/
e.	*Ø ₉ -kinini	‘pill’	French, /kinin/

To resolve this issue, I follow Aissen’s (1999) proposal for a constraint *Ø, which penalizes null morphology. This constraint states that morphological categories must be overtly realized.

(3) *Ø

No null morphology

I account for Kirundi’s preference to utilize existing phonological material for class prefixes with the constraint *Ø. FAITH-ALIGN-SB must outrank *Ø to explain why violations of *Ø are licensed to avoid violations of FAITH-ALIGN-SB, as demonstrated in Table 6 and Table 7. If *Ø were higher, we would expect every loanword to have an overt class prefix.

4.2 Consonant and vowel alternation

Compare the attested forms in Table 8 (I) and (II) with the ill-formed candidates in Table 8 (III) and (IV).

To avoid the potential problematic candidates in Table 8 (III), it is crucial to determine the ranking of faithfulness constraints relative to the aforementioned

Table 6: Evaluation of /kilo/

/kilo/	FAITH-ALIGN-SB	*Ø
☞ a. ki ₇ -ro		
b. Ø ₉ -kiro		*!
c. mu ₃ -kiro	*!	
d. ki ₇ -kiro	*!	

Table 7: Evaluation of /bætəi/

/bætəi/	FAITH-ALIGN-SB	*Ø
☞ a. Ø ₉ -βatiri		*
b. ki ₇ -βatiri	*!	
c. mu ₃ -βatiri	*!	

Table 8: Attested and unattested candidates

	Kirundi	Gloss	Source
I.	a. Ø ₉ -farine	‘flour’	French, /farin/
	b. Ø ₉ -pine	‘tire’	French, /pnØ/
	c. Ø ₉ -fa:ta	‘Fanta’	English, /fænta/
II.	a. *ma ₆ -rine	‘flour’	French, /farin/
	b. *βi ₈ -ne	‘tire’	French, /pnØ/
	c. *ma ₆ -:ta	‘Fanta’	English, /fænta/
III.	a. mu ₃ -dari	‘medal’	French, /mədajə/
	b. mu ₃ -sigiti	‘mosque’	Arabic, /masdʒid/
	c. mu ₃ -doka	‘car’	English, /mɔrækɑ/
	e. Ø ₉ -kurudzeti	‘zucchini’	French, /kurʒet/
	d. Ø ₉ -ko ⁿ dʒe	vacation	French, /kõʒe/
IV.	a. *ki ₇ -rudzeti	‘zucchini’	French, /kurʒet/
	b. *ki ₇ - ⁿ dʒe	vacation	French, /kõʒe/

FAITH-ALIGN-SB and *Ø. Kirundi loanwords could conceivably alter some segments in the input for it to resemble a class prefix. Indeed, forms such as those in Table 8 (II, a-c) indicate this happens with certain vowels. This does not happen to all vowels and does not happen for any consonants, however. There is no evidence that rounding is a contrastive feature in Kirundi vowels, so it is nonessential for our analysis. I propose the constraints IDENTV(BK)-SB, IDENTV(HI)-SB, and IDENTC-SB, for back and high vowels and consonants, respectively.

- (4) a. **IDENTV(BK)-SB**
Do not change the backness of any vowels from the source language in the base language
- b. **IDENTV(HI)-SB**
Do not change the height of any vowels from the source language in the base language
- c. **IDENTC-SB**
Do not change any consonants from the source language in the base language

As the consonant never changes to match a class prefix, IDENTC-SB must outrank both FAITH-ALIGN-SB and *Ø. Given that vowel quality can change to match some existing class prefix, some vowel faithfulness constraint should be ranked lower than FAITH-ALIGN-SB and *Ø. In Table 8 (II), we see only vowel height change and not backness, suggesting *Ø outranks IDENTV(BK)-SB. This is demonstrated in the tableau below.

Table 9: Evaluation of /kilo/

/kilo/	IDENTC	FAITH-ALIGN	*Ø	IDENTV(BK)
☞ a. Ø ₉ -kiro			*	
b. ma ₆ -kiro	*!			
c. βa ₂ -kiro	*!			

However, we do find some forms with variation. For example, (a) and (b) in Table 10 show it is possible to change the vowel to fit some existing class prefix, but also acceptable to keep a faithful vowel while adding covert class prefixes. We can stipulate that there is some level of variable ranking between *Ø and IDENTV(HI) amongst speakers. This would mean that for different speakers (and possibly different varieties), either candidate in Table 10 may be optimal. The tableau in Table 11 shows that both possible forms are attested.

Table 10: Variation in forms

	Kirundi	Gloss	Source
a.	Ø ₉ -modoka	‘car’	English, /motəkaɪ/
b.	mu ₃ -doka	‘car’	English, /motəkaɪ/

Table 11: Evaluation of /motəkaɪ/

/motəkaɪ/	IDENTC-SB	FAITH-ALIGN-SV	*Ø	IDENTV(HI)-SB
☞ a. mu ₃ -doka				*
☞ b. Ø ₉ -modoka			*	
c. ki ₇ -modoka		*!		
d. βu ₁₄ -modoka	*!			

I account for the observed asymmetry between back and front vowels by distinguishing IDENTV(BK)-SB and IDENTV(HI)-SB. In Table 8 (II, a-c), back vowels can change in height to resemble a class prefix. Examples in Table 8 (II, d-e) and Table 8 (IV, a-b) show that back vowels cannot be fronted for the same result. Thus, IDENTV(BK)-SB outranks at least *Ø, as demonstrated in the tableaux below.

Table 12: Evaluation of /kurzɛt/

/kurzɛt/	IDENTC-SB	FAITH-ALIGN-SV	*Ø	IDENTV(HI)-SB
☞ a. Ø ₉ -kurudzɛti			*	
b. ki ₇ -rudzɛti		*!		

4.3 Epenthetic vowels

Kirundi loanwords often use vowel epenthesis to break up complex syllable sequences (Niyondagara 1993). Niyondagara (1993) claims that the Kirundi syllable structure is best characterized as (C)V(V), the second V being an additional mora in long vowels. Kirundi also allows glides to follow consonants in onsets. Table 13 shows examples of vowel epenthesis breaking up syllable clusters.

Epenthetic vowels in Kirundi can appear in several potential forms. The two most common strategies involve copying an adjacent vowel or using the epenthetic vowel [i]. Example (a) in Table 13 shows the epenthesized vowel [e]

Table 13: Vowel epenthesis

	Kirundi	Gloss	Source
a.	Ø ₉ -petero:ri	‘gasoline’	French, /pɛtrɔl/
b.	Ø ₉ -po:sita	‘post office’	French, /pɔstə/

as a copy of the previous syllable’s nucleus. Kirundi loanwords show that the copied vowel can be on the left or right side of the epenthetic vowel. The other strategy, shown in (b) in Table 13, uses an epenthetic vowel [i]. Niyondagara (1993) notes that the most common epenthetic vowel in Kirundi is [i] and is the form seen when the consonant that would form the potential onset of a newly created syllable is /s/. We also see some instances in Kirundi where the epenthetic vowel is variable, giving two possible forms derived from the same source word. This is seen in (a-b) in Table 14, both acceptable and attested forms. Thus, we must account for (a) and (b) in Table 14 being variable. I deal with this variability in Section 5.

Table 14: Vowel epenthesis

	Kirundi	Gloss	Source
a.	βu ₁₄ -re ⁿ geti	‘blanket’	English, /bleɪnkɪt/
b.	βi ₈ -re ⁿ geti	‘blanket’	English, /bleɪnkɪt/

I refer to Rose & Demuth (2006)’s constraints and ranking to account for epenthesis in Kirundi loanwords. Rose & Demuth focus on Sesotho consonant cluster resolution, but I briefly show that their analysis can be successfully applied to Kirundi with minor modifications. The relevant constraints and ranking they suggest for Sesotho are:

- (5) **OK(σ) » CRISPEDGE(CV, σ) » DEP(VPL) » AGR_L(VPL) » AGR_L(CPL) » AGR_R(VPL)**

DEP(VPL) states that no new vowel place features may be added. In other words, if a vowel is added to the source word’s input, it should avoid new features by using the same features of some adjacent segment. AGR_L(VPL) and AGR_R(VPL) are constraints indicating which side’s vowel the epenthetic position is a copy of. AGR_L(CPL) states that segments should agree with consonants to the left. CRISPEDGE(CV, σ) (shown in tableaux as CE((CV, σ)) prevents consonant-vowel

agreement across syllable boundaries. Thus vowels only agree with the features of consonants in their syllable's onset. These constraints are given in (6a-6f) and the success of this ranking is shown in the tableaux in Table 15.

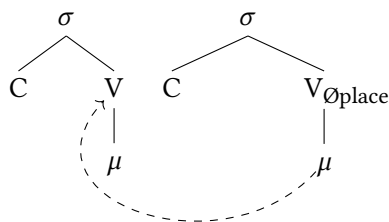
- (6) a. **DEP(VPL)** (Rose & Demuth 2006)
Every output VPlace specification has an input correspondent
- b. **AGRL(VPL)**
The place feature of the epenthetic vowel must agree with the place feature of the vowel immediately to its left.
- c. **AGRL(VPL)**
The place feature of the epenthetic vowel must agree with the place feature of the vowel immediately to its left.
- d. **AGRL(CPL)**
The place feature of the epenthetic vowel must agree with the place feature of the consonant immediately to its left.
- e. **CRISPEGE(CV, σ)**
Consonant–vowel feature sharing cannot occur across syllables.
- f. **OK(σ)**
Portmanteau constraint ensuring syllable well-formedness in output (adapted) forms

Table 15: Evaluation of /petrol/

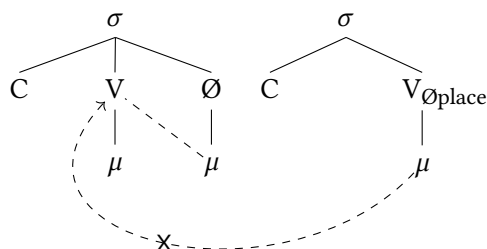
/petrol/	OK σ	CE(CV, σ)	DEP(VPL)	AGRL(VPL)	AGRL(CPL)	AGRR(VPL)
a. Ø ₉ -petero:ri						*
b. Ø ₉ -petro:ri	*!					
c. Ø ₉ -petoro:ri				*!	*	
d. Ø ₉ -petiro:ri				*!		*
e. Ø ₉ -petaro:ri			*!	*	*	*
f. Ø ₉ -peturo:ri			*!	*	*	*

Rose & Demuth (2006) do not discuss the role of vowel length in epenthetic forms. From the data, Kirundi epenthetic vowels do not seem to agree with long vowels to their left. To understand this, I adopt the theory of syllable structure and agreement relations shown in (7). In this structure, long vowels consist of two morae, the latter floating without being inherently associated to a vowel but becoming linked to the preceding segment (Zec 1995).

(7) a.



b.



The structure in (7a) shows an acceptable agreement relationship between two vowels. The structure in (7b) shows an ill-formed agree relationship between two vowels. By understanding long vowels as a vowel associated with two morae, we add the boundary of an additional mora that is not inherently associated with a vowel between the epenthetic site and the preceding vowel. This unspecified mora blocks the epenthetic vowel from probing backwards for a vowel to copy its place features from. Agreement between the epenthetic vowel and the left adjacent vowel is no longer possible, and a new agreement target is sought. This evaluation is shown in Table 16. No candidates trigger a violation of AGR_L(VPL) as the input does not see any vowel immediately left of the epenthetic site⁶.

Table 16: Evaluation of /pɔstə/

/pɔstə/	OK _σ	CE(CV,σ)	DEP(VPL)	AGR _L (VPL)	AGR _L (CPL)	AGR _{RR} (VPL)
☞ a. Ø ₉ -po:sita						*
b. Ø ₉ -po:sta	*!					
c. Ø ₉ -po:sata				*!	*	
d. Ø ₉ -po:sota				*!		*

⁶I leave the augment in tableaux to make the noun class more explicit, but it can be omitted and should not be seen as a potential goal for agreement.

Table 17

Kirundi	Gloss	Source
<i>mu</i> ₁ -perezida	‘president’	French, /prezidə/

I propose a final constraint to account for forms such as those in Table 17. The class prefix *mu*- is not a potential goal for vowel agreement, avoiding the form [mupurezida]. I propose the constraint *AGR MORPH BD(PL). This rules out the epenthetic vowel agreeing with the noun class prefix vowel, shown in the tableaux in Table 18. This constraint outranks AGR L(VPL), but its relative ranking to lower constraints is not crucial.

(8) *AGR MORPH BD(PL)

Do not agree with the place of segments across morpheme boundaries.

Table 18: Evaluation of /prezidə/

/prezidə/	OK σ	CE(CV, σ)	*AGR Mo Bd(PL)	DEP(VPL)	AGRL(VPL)	AGRL(CPL)	AGRR(VPL)
a. <i>mu</i> ₉ -perezida					*	*	
b. <i>mu</i> ₉ -prezida	*!						
c. <i>mu</i> ₉ -purezida							

5 Semantic factors in noun class allocation

Kirundi noun class allocation of loanwords is additionally conditioned by various semantic factors. Although Bantu noun classes are rarely entirely semantically coherent, we see strong restrictions and general tendencies affecting a noun’s class allocation. Thus, we cannot account for the distribution of noun class prefixes to loanwords with phonological criteria alone.

In this section, I outline notable patterns in the semantic effects of noun class allocation in Kirundi loanwords. In Section 6, I propose an analysis reconciling the conflict between semantics and phonology, demonstrating a capability for one unified interface.

5.1 Animacy

I first view the effect of animacy on Kirundi noun class allocation for a simple reason: [$\pm$ human] is the only semantic feature that correlates perfectly with a

particular noun class. Zorc & Nibagwire (2007; among others) note that nouns of class 1/2 always denote human entities. Not all human nouns belong in class 1/2, but every noun in this class is a human.

There is a general tendency for Kirundi loanwords to be allocated to class 1/2 when denoting a human. Often, this involves adding the class 1 prefix to an existing input to mark that noun as belonging to that noun class, as seen in Table 19. In (a) in Table 19, the overt class 1 prefix, /umu-/ is added to the source word.

In other cases, loanwords with initial segment sequences resembling a class 1/2 prefix may have their initial CV sequence repurposed when denoting human nouns. In (d) in Table 19, the loanword /mus.ləm/ is incorporated to the language as /mu₁-silamu/, using phonological material from the source word for the class 1 prefix.

Table 19

	Kirundi	Gloss	Source
a.	mu ₁ -perezida	'president'	French, /prezidə/
b.	mu ₁ -furatiri	'brother'	Latin, /frater/
c.	mu ₁ -kirisito	'christian'	Christian
d.	mu ₁ -silamu	'muslim'	Arabic, /mʊs.ləm/
e.	mu ₁ -se:neri	'mister'	French, /məsjø/

Because class 1/2 is reserved solely for human nouns, loanwords with initial segment sequences resembling a class 1/2 prefix are put into other classes. This is shown in (a) in Table 20, where the homophonous class 3 prefix is used instead of the class 1 prefix. In (20b), the class 2 prefix is avoided.

Table 20

	Kirundi	Gloss	Source
a.	mu ₃ -sigiti	'mosque'	Arabic, /mas.dʒid/
b.	Ø ₁₀ -βata	'duck'	Swahili, /bata/

We can establish that those nouns in Table 19 are given class 1 prefixes rather than the class 3 prefixes on the basis of noun class agreement with other elements in the sentence and their non-singular noun class alternation.

- (9) a. U-mu-se:neri a-fis-e a-βa-ána be-nshi.
 A-CL1-mister 1SM-have-PFV A-CL2-child CL2-many
 ‘The man has many children.’
 b. A-βa-se:neri ba-fis-e a-βa-ána be-nshi.
 A-CL2-mister 2SM-have-PFV A-CL2-child CL2-many
 ‘The men have many children.’
- (10) a. U-mu-sigiti mu-fis-e a-βa-ntu be-nshi.
 A-CL3-mosque 3SM-have-PFV A-CL2-person CL2-many
 ‘The mosque has many people.’
 b. I-mi-sigiti i-fis-e a-βa-ntu be-nshi.
 A-CL4-mosque 4SM-have-PFV A-CL2-person CL2-many
 ‘The mosques have many people.’

In (9a), *umuse:neri* triggers class 1 verbal subject agreement. In (9b), the same noun is given a class 2 prefix, triggering class 2 subject agreement on the verb. In (10a), *umusigiti* triggers class 3 subject agreement on the verb and alternates with the class 4 prefix when plural in (10b).

5.2 Cumulativity

Another mostly semantically coherent aspect of noun class is cumulativity. Oliver & Kinchsular (2022) note that even-numbered noun classes in Kirundi all denote entities with cumulative reference⁷. Only pluralities or non-atomic entities are found in these classes. Class 6 is often where substance-denoting nouns are found and is understood as a ‘polyplural’ class in Bantu languages (Maho 2009, Zorc & Nibagwire 2007, Oliver & Kinchsular 2022). Cumulative reference is shared by individuated plural nouns and non-individuated/mass nouns.

Instances where nouns with cumulative reference have initial segment sequences reanalyzed as noun class prefixes are seen in Table 21 (I). Other noun classes beyond class 6 may also show cumulative reference. There are also instances where loanwords with sounds resembling a non-individuated noun class prefix may not be interpreted as those class prefixes when lacking cumulative reference. Instead, these nouns either take an overt prefix or take the covert class 9 prefix. This is seen in Table 21 (II). Unlike human-denoting nouns allocated to class 1, we do not see instances of nouns with cumulative reference being automatically allocated to class 6.

⁷Noun classes above 10 are exceptions to this rule. However, this is a consequence of them being derivational classes and thus should not be understood as regular in regard to the

Table 21

		Kirundi	Gloss	Source
I.	a.	ma ₆ -joneze	'mayonnaise'	French, /mejones/
	b.	Ø ₁₀ -βata	'ducks'	Swahili, /bata/
II.	a.	Ø ₉ -mandata	'mandate'	French, /māda/
	b.	Ø ₉ -mafi:ne	'machine'	French, /mafin/

5.3 Abstract and diminutive nouns

Less semantically coherent but still salient semantic categories are those of abstract nouns and diminutives. Zorc & Nibagwire (2007) observe that class 14 is largely comprised of abstract nouns, while classes 12 and 13 are primarily nouns with diminutive meanings.

- (11) a. i-gi-tabo
A-CL7-book
'book'
- b. a-ga-tabo
A-CL12-book
'pamphlet'

6 Phonosemantic interface

In this section, I use an optimality theoretic approach, with crucially ranked phonological and semantic constraints to account for the interaction of phonological and semantic factors. Kirundi's noun classes are not usually fully semantically cohesive. However, an optimality theoretic ranking system is well-equipped to handle the lack of complete semantic cohesion, allowing for violations of noun class constraints to prevent a violation of some higher-ranked constraint.

In sections 6.1-6.3, I propose constraints and crucial rankings that account for the semantic factors described above. In 7, I consider and argue against an alternative analysis where a division between the semantic and phonological interfaces is upheld.

individuated/non-individuated contrast.

6.1 Human constraints

I first propose a constraint to derive the human semantics of class 1/2. To capture that human nouns are strongly preferred in class 1/2 and that only human nouns can comprise class 1/2, I use the following constraint:

(12) **HUMAN**↔1/2

All and only human nouns in class 1/2.

As noted in 5.1, there are instances where human-denoting nouns beginning with segments resembling class 1/2 prefixes may have those segments reinterpreted as class 1/2 prefixes. Some examples of this are repeated in Table 22.

Table 22

	Kirundi	Gloss	Source
a.	mu ₁ -silamu	‘Muslim’	Arabic, /mʊs.ləm/
b.	mu ₁ -se:neri	‘mister’	French, /məsʝØ/

I have also noted cases where human-denoting nouns without any segments that resemble class 1/2 prefixes can have an overt prefix assigned. This happens when human denoting loanwords have no initial sequence resembling a class 1/2 prefix. Examples are given in Table 23.

Table 23

	Kirundi	Gloss	Source
a.	mu ₁ -perezida	‘president’	French, /prezidə/
b.	mu ₁ -furatiri	‘brother’	Latin, /frater/
c.	mu ₁ -kirisito	‘christian’	Christian

It becomes clear from this evidence that our semantic constraint must interact with the previous constraints proposed. The preference to have human nouns in class 1/2 apparently takes precedent over the preference for loanwords to appear in class 9/10 without disrupting word boundaries. To capture this, we need only rank **HUMAN**↔1/2 higher than * $\emptyset$ and **FAITH-ALIGN-SB**. In [umuperezida], for example, a violation of **FAITH-ALIGN-SB** is incurred to avoid triggering a potential violation of **HUMAN**↔1/2 that would arise without assigning the noun to class 1/2. If **FAITH-ALIGN-SB** were ranked higher, we would expect the noun to be allocated to class 9/10 with a covert noun class marker.

Table 24: Evaluation of /prezidə/

/prezidə/	HUM↔1/2	IDENTC-SB	*Ø	FAITHALIGN-SB	IDENTV-SB
☞ a. mu ₉ -perezida				*	
b. Ø ₉ -perezida	*!		*		

6.2 Cumulative constraints

To contain the discussion of correlation between even numbered and cumulative noun classes, I only consider class 6 loanwords. Class 6 houses many substance-denoting nouns and serves as the irregular plural for many. I introduce the following constraint to account for this property of class 6;

(13) CUMULATIVE↔6

All and only nouns with cumulative reference are allocated to class 6

Not all substance-denoting loanwords are in class 6, as seen in Table 25, where the substance-denoting noun [isukari] ('sugar') is realized with a class 10 prefix. A possible alternative would have this noun be realized as [*amasukari]. That [isukari] is the preferred form suggests that the constraint penalizing candidates with unfaithful word boundaries outranks our semantic constraint for class 6.

Table 25: Evaluation of /sykr/

/sykr/	IDENTC-SB	FAITHALIGN-SB	CUMUL↔6	*Ø	IDENTV-SB
☞ a. Ø ₉ -sukari			*	*	
b. mu ₉ -sukari		*!			

Cumulative-reference nouns with initial segments resembling the class 6 prefix can have initial segments repurposed as a class prefix. Nouns without cumulative reference resembling the class 6 prefix cannot, as seen in Table 22. This suggests that CUMULATIVE↔6 outranks *Ø, as violations of *Ø can occur to avoid violations of CUMULATIVE↔6. The difference is shown in Table 26 and Table 27.

Table 26: Evaluation of /mejonesə/

/mejonesə/	IDENTC-SB	FAITHALIGN-SB	CUMUL↔6	*Ø	IDENTV-SB
☞ a. ma ₆ -joneze					*
b. Ø ₉ -mejoneze			*!	*	

Table 27: Evaluation of /mafinə/

/mafinə/	IDENTC-SB	FAITHALIGN-SB	CUMUL↔6	*Ø	IDENTV-SB
 a. Ø ₉ -mafine				*	
b. ma ₆ -fine			*!		

6.3 Abstract and diminutive constraints

I now consider some of the derivational noun classes in Kirundi. Classes 12/13 and 14 behave differently from other Kirundi noun classes. Class 12/13 is the diminutive noun class. As demonstrated previously in (11), nouns in class 12/13 are given diminutive interpretations. Class 14 is mostly reserved for abstract nouns. Abstract qualities can be expressed by replacing the inherent class prefix of other roots with the class 14 prefix.

The number of nouns inherently belonging to either class is highly limited and are often lexicalized nouns that can be traced back to some derived form (Zorc & Nibagwire 2007). I discuss these noun classes as their derivation use makes them clearly semantically cohesive. Following similar treatment for previous noun classes considered, I introduce the following two constraints to account for these classes' semantic cohesion:

- (14) **DIMINUTIVE↔12/13**
All and only diminutive nouns in class 12/13
- (15) **ABSTRACT↔14**
All and only abstract nouns in class 14

We have examples of loanwords that begin with sequences of segments resembling class 12/13 or class 14 prefixes that are not semantically cohesiveness with other nouns in those classes. With [imaʃi:ne], the semantic constraint overrides the phonological constraint to ensure the noun is not inappropriately allocated to class 6. We see the opposite with class 12/13 and 14 loanwords, where semantic constraints are clearly violated. For example, in Table 28, [akaβati] ('cabinet') is allocated to class 12 without having any diminutive sense, and [uβu₁₄reⁿgeti] ('blanket') is allocated to class 14, despite being a concrete noun. The semantic constraints for class 12/13 and class 14 therefore must be outranked by phonological constraints. This is shown in Table 29 and Table 30.

Semantic and phonological constraints relevant to noun class allocation can vary in ranking, with the hierarchy in (16). This further supports the proposal that the semantic and phonology interface work as one in noun class allocation.

Table 28

	Kirundi	Gloss	Source
a.	ka ₁₂ -βati	‘cabinet’	English, /kæbnɪt/
b.	ka ₁₂ -βare	‘cabaret’	English, /kæbəreɪ/
c.	βu ₁₄ -re ⁿ geti	‘blanket’	English, /blemkɪt/

Table 29: Evaluation of /blemkɪt/

/blemkɪt/	OK(σ)	FAITHALIGN-SB	*Ø	AGRL(CPL)	ABST↔14	DIM↔12/13
☞ a. βu ₁₄ -βure ⁿ geti				*		
b. Ø ₉ -βure ⁿ geti			*!			
c. ki ₇ -re ⁿ geti		*!				

(16) HUMAN↔1/2, IDENTC-SB » FAITHALIGN-SB » CUMULATIVE↔6 » *Ø »
ABSTRACT↔14, DIMINUTIVE↔12/13

Considering the abstract class 14, we also gain insight into the relative ranking of semantic and phonological constraints determining vowel epenthesis. As noted in Section 3, [uβu₁₄reⁿgeti] and [iβi₈reⁿgeti] are both accepted forms in Kirundi. As a vowel must be epenthesized to resolve the consonant cluster in ‘blanket’, the relevant phonological constraint for determining the form of that vowel is AGRL(CPL). In [uβu₁₄reⁿgeti], the epenthetic vowel is rounded to agree with the consonant, but [iβi₈reⁿgeti] triggers a violation of AGRL(CPL). Considering semantic constraints, [uβu₁₄reⁿgeti] incurs a violation of ABSTRACT↔14, as the noun is not abstract in nature, while [iβi₈reⁿgeti] avoids violating the constraint by allocating the noun to a different class.

Accounting for variation between these candidates, we can assert there is no critical ranking between AGRL(CPL) and ABSTRACT↔14. This is demonstrated in Table 31.

Table 30: Evaluation of /kæbnɪt/

/kæbnɪt/	OK(σ)	FAITHALIGN-SB	*Ø	ABST↔14	DIM↔12/13
☞ a. ka ₁₄ -βati					*
b. Ø ₉ -kaβati			*!		

Table 31: Evaluation of /blemkit/

/blemkit/	OK(σ)	HUM $\leftrightarrow$ 1/2	FAITHALIGN-SB	* $\emptyset$	AGRL(VPL)	ABST $\leftrightarrow$ 14	AGRL(CPL)
 a. βu_{14} - βure^n geti						*	
b. β_8 -re n geti							*
c. $\emptyset_9$ - βure^n geti		*!		*			
d. $\emptyset_9$ - βre^n geti	*!						
e. βu_2 -re n geti		*!					
f. β_8 - βure^n geti			*!				

7 Interface implications

In the preceding sections, I have argued that semantics and phonology can be construed as using one and the same interface. Using an Optimality Theory framework, I have demonstrated that semantic and phonological factors and constraints can be ranked in relation to each other in the same way that simple phonological constraints can. I consider an alternative analysis for the data presented. An alternate analysis may posit that phonology and semantics do not interact directly, but one ‘feeds’ the other, giving it input for evaluation. I evaluate the shortcomings of this approach and show that a unified phonosemantic interface is favourable. I return to [imaʃi:ne] as an example to illustrate an important behaviour lending support to a unified system.

If we consider the analysis where phonology feeds the semantic interface, we could explain much of the data observed as simple cases of derivation. In such a hypothetical analysis, we might assume that loanwords are first allocated to a particular noun class based on phonological principles. Once allocated, some semantic evaluation system determines whether that loanword class allocation is acceptable based on its lexical content. If deemed acceptable, the loanword maintains the given class prefix. If unacceptable, the noun is reallocated to a more suitable noun class. Crucially, this would involve phonology incorporating the noun into the morphological system before the semantic interface could evaluate the noun class allocation. However, I argue this cannot be the case, and phonology and semantic considerations must work in tandem to determine noun class allocation. Consider how such an account would apply to the loanword [imaʃi:ne].

In [imaʃi:ne], the sequence resembling the class 6 prefix is retained, and the source word is simply augmented with a class 9 prefix. The approach described above cannot account for this behaviour. We must realize that both semantic and phonological constraints can be violated in favour of the other.

If there were indeed two distinct interfaces, we would expect [*iʃi:ne]. The phonological interface could first analyze the existing sequence of segments in

/ma₉finə/. After the appropriate sound changes occur, we would expect the morphophonological form to enter the Kirundi lexicon as /ma₆-fi:ne/. The tableaux in Table 32 and Table 33 demonstrate EVAL within this approach, where semantic constraints do not interact with phonology. This would separate the loanword into root and affix before any semantic factors are considered. The semantic interface could then evaluate the lexical item [ma₆-fi:ne], determine that class 6 is not a semantically appropriate class for the item in question, and replace the prefix with any prefix that does not indicate cumulative reference. This could give us the form [Ø₉-fi:ne], to which the augment would be added afterward.

Table 32: Phonology Eval

/ma ₉ finə/	IDENTC-SB	FAITHALIGN-SB	*Ø	IDENTV-SB
☞ a. Ø ₉ -ma ₉ fine			*!	
b. ma ₆ -fine				

Table 33: Semantic Eval

/ma ₉ finə/	HUMAN↔1/2	CUMULATIVE↔6
☞ a. Ø ₅ -fine		
b. ma ₆ -fine		*!
c. mu ₁ -fine	*!	

The forms we see suggest that specific phonological and semantic constraints can be overridden by the other. The only way to properly account for these forms is by concluding one of two things. I propose that semantics and phonology either have the same interface, where semantic and phonological constraints are crucially ranked with respect to each other, or the distinct interfaces are intertwined to the degree that it becomes undesirable and ineffective to view them separately.

8 Conclusion

In the preceding sections, I hope to have shown an analysis of Kirundi loanwords that demonstrates a unity between semantics and phonology is possible within an optimality theory framework. I have focused on how semantic and phonological factors contribute to the allocation of Kirundi loanwords into the rich noun class system.

In Section 3 and Section 4, I first focused on phonological patterns in Kirundi noun class allocation of loanwords. The tendency for loanwords in Kirundi to be allocated primarily to noun classes with null prefix morphemes or to use existing phonological material in the source language can be understood as primarily the result of constraints seeking to maintain faithful word boundaries from source to the base language and constraints requiring that morphology must be overt. These constraints, in addition to other faithfulness constraints penalizing the alteration of segments, can fairly reliably predict the designation of Kirundi loanwords to the noun class system.

In addition, epenthesis and syllable structure constraints interact with these faithfulness constraints. The restrictions set on well-formed syllables in Kirundi often lead loanwords to require some sort of syllable restructuring. In the process of incorporating these loanwords into the noun class system, Kirundi can exploit some syllabification processes to ensure that the output fits well within the noun class system.

In Section 5, I showed that semantic conditioning and noun class restrictions factor into determining how a loanword is assigned a noun class. Semantic factors of the loanword can cause irregularities in the phonological patterns of noun class allocation described in the previous section. In Section 6, I addressed this by providing an account of Kirundi loanword allocation that takes both semantic and phonological factors into consideration. The analysis I proposed is one where phonology and semantic conditioning of noun classes essentially play different roles in the same system.

An asymmetric ranking of semantic noun class constraints within an optimality theoretic approach accounts for the observed asymmetry in the semantic cohesion of Kirundi noun classes. Therefore, certain tendencies of Kirundi noun classes to be stronger than others are explained by constraints that may be violated to avoid violating higher-ranked constraints, whether semantic or phonological. There are numerous instances both of higher-ranked semantic constraints leading to the violation of some phonological constraints and the violation of lower semantic constraints incurred by avoiding the violation of phonological constraints.

This paper is by no means a conclusive study of the phonology of Kirundi noun class prefixes or the semantic factors of Kirundi noun classes. I hope to have made some small contribution towards an understanding that semantics and phonology can be interconnected to a greater degree than is regularly granted. The analysis I have proposed lends credence to an argument that semantic and phonological patterns can be accounted for within the same system, and there

is a clear benefit to doing such, as any alternative can not as effectively capture the interaction between the two.

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References

- Aissen, Judith. 1999. Markedness and subject choice in Optimality Theory. *Natural Language & Linguistic Theory* 17(4). 673–711.
- Aunio, Lotta Harjula. 2006. The Ha noun class system revisited. *SKY Journal of Linguistics* 19. 200–208.
- Cox, Elizabeth E. 1969. *Dictionary: Kirundi-English, English-Kirundi*. Winona Lake: General Missionary Board of the Free Methodist Church.
- Givón, Talmy. 1972. Studies in ChiBemba and Bantu grammar. *Studies in African linguistics* supplement 3. 1–247.
- Kayigema, Jacques. 2018. Allocation of loanwords into Kinyarwanda: Specific areas of influence. *Macrolinguistics* 6. 118–138. DOI: 10.26478/ja2018.6.8.8.
- Maho, Jouni. 2009. *NUGL online the online version of the new updated Guthrie list, a referential classification of the Bantu languages*. https://brill.com/fileasset/downloads_products/35125_Bantu-New-updated-Guthrie-List.pdf?srsId=AfmBOorILmO3Cs5xUtHR_s32uqm-b46eelzTdg1SK1O5AjEiBLtwUUIT.
- Mufwene, Salikoko S. 1980. Number, countability and markedness in Lingala LI/MA- noun class. *Linguistics* 18. 1019–1052.
- Ndayiragije, Juvénal, Emmanuel Nikiema & Parth Bhatt. 2012. The augment in Kirundi: When syntax meets phonology. In *Selected proceedings of the 42nd Annual Conference on African linguistics*, 112–121. Somerville.

- Niyondagara, Alice. 1993. *Kirundi phonological representations and the formation of complex segments*. Stony Brook: State University of New York. <https://books.google.ca/books?id=Je7cJwAACAAJ>.
- Niyonkuru, Lothaire. 1987. Loan-words in Kirundi: A preliminary study. *African study monographs* 7. 81–87.
- Oliver, Mikayla & Patrick Kinchular. 2022. Examining the mass/count distinction in Kirundi and Tshiluba. In *Presentation at the 15th annual Toronto Undergraduate Linguistics Conference*.
- Rose, Yvan & Katherine Demuth. 2006. Vowel epenthesis in loanword adaptation: Representational and phonetic considerations. *Lingua* 116. 1112–1139. DOI: 10.1016/j.lingua.2005.06.011.
- Rwamo, Alice & Ntiranyibagira Constantin. 2019. Phonological and perceptual factor symbiosis in loanword adaptation. *Revista Odisseia* 5. 22–39. DOI: 10.21680/1983-2435.2020v5n1ID18827.
- Shanks, David. 2023. *Spelling out the Kirundi augment: Prosodic domains, epenthesis and weight* (Toronto Working Papers in Linguistics 46).
- Van de Velde, Mark. 2019. Nominal morphology and syntax. In Koen Bostoen, Mark Van de Velde, Derek Nurse & Gérard Philippson (eds.), *The Bantu languages*, 2nd edn., 237–269. London: Routledge.
- Zec, Draga. 1995. The role of moraic structure in the distribution of segments within syllables. In Jacques Durand & Francis X. Katamba (eds.), *Frontiers of phonology: Atoms, structures, derivations*, 149–179. London: Longman.
- Zorc, R. David & Louise Nibagwire. 2007. *Kinyarwanda and Kirundi comparative grammar*. Hyattsville: Dunwoody Press.